

The clinical-language intelligence layer for behavioral health.

Vasl reads the language youth actually use — in clinical notes, screeners, coaching, and peer community — and surfaces distress signals for a clinician's independent review. Non-device clinical decision support, built for the settings where young people already show up.

CONTENTS

- 01 The gap this closes
- 02 What Vasl is
- 03 VLAP — the technology
- 04 How it functions as clinical decision support
- 05 The continuum of care
- 06 Deployment settings
- 07 Rural and Appalachian deployment
- 08 Language equity and the science behind it
- 09 Evidence, validation status, and starting a pilot

THE PROBLEM

The signal is already in the chart. Nobody reads it twice.

A young person in behavioral health distress arrives in the emergency department. They are stabilized, discharged, and handed a referral. The wait for a first therapy appointment runs six to nine months. Nothing happens in between, so they come back.

What gets missed

The intake note, the discharge summary, the free text on a screener, the message a 16-year-old sends a peer at 1 a.m. — that is where distress announces itself first. It sits in unstructured text that no downstream clinician opens again.

Structured risk fields capture what was asked. Language captures what was said.

Why general-purpose NLP under-reads it

Off-the-shelf clinical NLP is trained on text that does not sound like the youth being screened. Distress phrased in African American Language, in Appalachian English, in Spanish-influenced registers, or in ordinary teenage vernacular reads as noise to a model built on adult, standardized, majority-register writing.

The population most likely to be under-detected is the population most likely to return through the ED.

What changes with Vasl in place

The same encounter runs differently. Intake and discharge notes are analyzed and distress signals are surfaced for the clinician to review before the youth leaves. At discharge, the social worker enrolls them in a Vasl peer group. Support starts that day — not in the second quarter of next year.

OUTCOMES FROM VASL COMMUNITY PILOT PROGRAMS

79.5%

30-day retention in program

42%

PHQ-8 improvement at 90 days

6–9 mo

Typical wait for a first therapy appointment — the window Vasl fills

OVERVIEW

Infrastructure, not an app.

A Delaware public benefit corporation building behavioral health infrastructure for underserved, BIPOC, LGBTQIA+, and limited-English-proficiency youth. Two things make up the company: an engine that reads clinical language, and the care surfaces that engine feeds.

LAYER ONE

VLAP

The Vasl Language Analysis Platform. A fine-tuned transformer that reads free text from any capture surface and surfaces 47 clinical signals across six categories for clinician review. Vasl's own model, own weights.

LAYER TWO

The care surfaces

Five role-scoped portals plus an API, delivering stepped care: peer community, coaching, and routed referral into the partner's own therapy and crisis pathways. Every surface generates language, and every conversation makes the engine better at reading the next one.

Who sees what

Access is scope, not seniority. Signal detail stops at the clinician boundary by design — a coach never sees a score, a member never sees a clinical label.

ROLE	SURFACE	WHAT THEY CAN SEE
Member	app.gotovasl.com	Their own experience. No scores and no clinical language, ever.
Coach	coach.gotovasl.com	Engagement metadata only. Whether she finished the check-in, not what she answered.
Clinician	clinician.gotovasl.com	Credentialed, full clinical read: surfaced signals, the language behind them, notes.
Org admin	admin.gotovasl.com	Widest role inside one organization — identified members, notes, surfacing tier, audit, access.
Super admin	super.gotovasl.com	Every tenant, de-identified aggregates only. Cannot resolve a member to a name.

Where Vasl deploys

Health systems, federally qualified health centers, school-based health centers, schools, YMCAs, libraries, and community health organizations.

How it runs

Distributed services on AWS under an active business associate agreement. Integration by HL7 FHIR R4 — Epic, Cerner, athenahealth — or flat CSV/JSON where FHIR is not available.

THE ENGINE

Vasl Language Analysis Platform

A transformer-based natural language engine trained to read distress in the registers young people actually speak, and to hand a clinician the specific words that triggered every flag.

Taxonomy

47 signals across **six clinical categories** — HOP, SHA, CCM, ISO, TRM, PFE. Canonical taxonomy VASL-VLAP-TAX-001 v1.0; full definitions available under NDA.

Model

DeBERTa-v3-base fine-tuned on Vasl corpora. Vasl's own weights on Vasl's own infrastructure — not a prompt wrapper around a general-purpose LLM.

Interface

Versioned REST service at `/api/vlap/v2/*`, deployed as an independent service. Text in, signals plus the evidence spans out.

Any capture surface

VLAP is capture-agnostic on purpose. The same engine reads whichever surface a partner already has, which is what makes it infrastructure rather than a feature.

SOURCE A

EHR clinical notes

Intake, progress, and discharge documentation, anchored as CDS in the clinician's workflow.

SOURCE B

Screeners free text

PHQ-8 and PHQ-A responses, including everything written outside the scored fields.

SOURCE C

Coaching interactions

Session notes and check-ins from the coaching layer of the care model.

SOURCE D

Peer community

Moderated member-to-member language — candid youth distress phrasing available nowhere else.

Output: signals, not verdicts

Each result returns the matched language, the signal, its category, and a per-signal risk level of High, Moderate, Low, or Variable. Signals are then routed by a **surfacing tier** — clear, monitor, high, or crisis. The tier is operational routing that sets review order and queue placement. It is not a clinical grade and is never presented as one.

Equity gating on every release

Model releases are scored across sociolect slices, not just on aggregate accuracy. A release that improves overall performance while widening dispersion across language varieties does not ship to member-facing surfaces. That gate has held back a release whose headline metric improved — which is the point of having it.

THE MOAT

Every conversation across every Vasl surface generates candid youth-distress language that does not exist in any public corpus or licensed clinical dataset. The engine improves with every token it reads, and the data that improves it cannot be bought.

REGULATORY POSTURE

Non-device clinical decision support.

VLAP is designed, documented, and operated as non-device CDS under section 3060 of the 21st Century Cures Act and FDA's 2022 clinical decision support software guidance. It is not offered as software as a medical device.

VLAP surfaces signals for a clinician's independent review. It never determines, diagnoses, decides, or escalates on its own.

How the four criteria are met

CRITERION	HOW VLAP SATISFIES IT
Not a signal or image analyzer	VLAP reads written clinical and member language. It does not acquire, process, or analyze a medical image, an in vitro diagnostic signal, or a pattern from a signal acquisition system.
Displays and analyzes medical information	Inputs are notes, screener responses, and member text already held by the care organization.
Supports a professional's judgment	Output is surfaced to credentialed clinicians as prioritized review material. No clinical action is taken by the software.
Basis is independently reviewable	Every flag ships with the exact language that produced it, the signal and category assigned, and the taxonomy definition behind them. The clinician can reach the same conclusion without relying on the software.

What VLAP does

- Surfaces signals in language a clinician might not reach in time
- Flags encounters for review and prioritizes a review queue
- Informs a clinician's assessment with the evidence attached
- Documents what was surfaced, to whom, and when

What VLAP does not do

- Diagnose, or assign a clinical risk score to a patient
- Escalate, alert externally, or act without a human
- Display clinical scores or diagnostic language to members or coaches
- Substitute for a validated instrument or a clinical interview

Governance and data handling

Role-based access is enforced at the router, not only in the interface, and tenant isolation is tested. Sensitive data is encrypted at rest. All hosting runs on AWS under an active business associate agreement, with an LLM provider that is itself BAA-eligible. Member text is never used to train third-party models. Access events against identified member data are audit-logged.

CARE MODEL

Therapy Last, Not Therapy Never.

Most youth in distress do not need a therapist first. They need something the same week. Vasl runs a stepped continuum where the lowest-intensity support starts immediately and intensity rises only when the language says it should.

STEP	WHAT HAPPENS	WHAT VLAP CONTRIBUTES
1 • Peer community	The youth joins a moderated peer group the day they enroll. No waitlist, no intake bottleneck, nothing clinical to opt into.	Reads the language of the community; surfaces signals for clinician review.
2 • Signals surfaced	Distress language across community, screeners, and notes is flagged and queued for clinician review by surfacing tier.	This is the step. Evidence spans attached to every flag.
3 • Coaching	An eight-week curriculum track with a trained coach. Check-ins, reminders, and progress against per-coach capacity.	Coach sees engagement metadata only — completion, progress, recency.
4 • Therapy referral	When the clinician determines therapy is warranted, Vasl routes the referral to the partner's preferred therapy provider. Not by default, and not by algorithm.	Signal history travels with the referral, so the receiving clinician has context from day one.
5 • Crisis referral	Human-directed referral into the partner's existing crisis pathway, under the partner's own protocol.	Surfaces the language. The human decides and acts.

Vasl does not provide therapy. Steps 1 through 3 are Vasl's. Step 4 routes to the partner's preferred therapy provider, and step 5 runs on the partner's own crisis protocol. Vasl employs no therapists and contracts none — the clinical tier stays inside the partner organization.

Why the order matters

Therapy-first models fail on supply. There are not enough clinicians, and the ones who exist are booked past the point where a referral means anything. Putting peer support and coaching first keeps the partner's clinical tier available for the youth who need it. It is also where the 79.5% retention comes from.

Where the health system plugs in

A partner can enter at any step. An emergency department typically enters at step 2 for note analysis and step 1 for discharge enrollment. An FQHC often enters at step 3, using coaching to hold patients who would otherwise sit on a referral list.

The discharge moment

The highest-leverage handoff in the model. Notes are analyzed and signals surfaced for the clinician before the youth leaves, and the social worker enrolls them in a peer group at discharge. The youth walks out with something that starts today, not just a referral with a two-quarter wait.

WHERE IT RUNS

One engine, three deployment patterns.

The clinical question changes by setting. What VLAP reads, who reviews it, and what the partner does with it change with it.

HEALTH SYSTEM

ED and inpatient behavioral health

Reads: intake notes, progress notes, discharge summaries, screener free text.

Reviewed by: ED clinicians, behavioral health consult service, social work.

Changes: distress language is surfaced before discharge rather than after readmission, and discharge becomes an enrollment point instead of a referral slip.

FQHC / SBHC

Primary care and school-based health

Reads: PHQ-8 free text, visit notes, coaching check-ins.

Reviewed by: the behavioral health integrationist or the supervising clinician.

Changes: the panel of adolescents who screen positive and then wait on an external referral gets held inside the coaching tier, with signals surfaced continuously instead of at the next annual visit.

COMMUNITY

Schools, YMCAs, libraries, youth orgs

Reads: peer community language and program check-ins.

Reviewed by: the partnered clinical supervisor; program staff see engagement only.

Changes: a non-clinical setting gains a clinical safety net without becoming a clinic. Pilot settings have included Thread YMCA, Enoch Pratt Library, the University of Maryland, the University of Baltimore, and Boys & Girls Clubs.

Integration

Data in

HL7 FHIR R4 against Epic, Cerner, or athenahealth, or flat CSV/JSON where FHIR is not available. Behavioral health encounters, structured and unstructured notes, ICD-10 for mood, anxiety, trauma and substance use, screening instruments, demographics and SDOH. De-identified to HIPAA Safe Harbor for research or validation use.

Output out

Signals and evidence spans delivered to the Vasl clinician portal, or returned through the API into the partner's own review surface. Nothing is written back into the chart without the partner's explicit configuration and a clinician in the loop.

A pilot in three phases

- 1 **Retrospective.** De-identified historical notes analyzed offline. No workflow change, no clinical reliance. Establishes signal yield and vocabulary fit against the partner's own documentation.
- 2 **Shadow.** Live text and live surfacing, reviewed alongside the existing process without depending on it. Establishes agreement and review burden.
- 3 **In workflow.** Surfaced signals enter the clinical review queue as CDS, with the continuum available at discharge or referral.

RURAL DEPLOYMENT

Rural is not a smaller version of urban.

A different language problem, worse workforce math, and money already appropriated. West Virginia is where Vasl is building the rural case.

THE WORKFORCE MATH

West Virginia carries **119 active mental health shortage-area designations**. HRSA puts the gap at **91 mental-health clinician FTEs** to clear them statewide.

You cannot hire your way out of that. A model that opens with peer support and coaching, reserving clinical hours for the youth who need them, is not a compromise here. It is the only configuration that runs.

THE FUNDING CHANNEL

West Virginia was awarded **\$199.5 million for 2026** under the federal Rural Health Transformation Program — year one of five, through 2030, administered by the West Virginia Secretary of Health.

State priorities name behavioral health treatment, addiction recovery, telehealth expansion, and workforce. Vasl sits inside all four.

PROCUREMENT-READY IN WEST VIRGINIA

Registered vendor

WV OASIS VSS, submitted September 4, 2026. Vendor code VS0000053719.

Federally registered

SAM.gov UEI RMBYZMMJAPP9. Self-attested minority-owned; not MBE-certified.

Service area

All 55 West Virginia counties.

Appalachian English is the same measurement problem

A rural sixteen-year-old describing hopelessness does not phrase it the way a clinical training corpus does, and the failure mode is the one Vasl was built to fix for African American Language: the model reads real distress as noise. Vasl's method is variety-general by construction — language variety is measured and carried as a covariate rather than corrected away, and every release is gated on dispersion across sociolect slices.

Telehealth-native by design

Peer community and coaching run remotely, so they reach a county with no behavioral health clinician in it. Therapy referrals route to the partner's own provider and the crisis pathway stays local and human under the partner's protocol. That is what makes a 55-county footprint a deployment question rather than a hiring question.

Where it lands: critical access hospital EDs, FQHCs, rural health clinics, school-based health centers, county schools.

What an RHTP partnership would build

Vasl has a clinically annotated corpus for African American Language. It has none for Appalachian English — no one does, which is exactly the argument for funding it. It would be built the same way: community annotators with dialect and lived competence, independent paid raters, clinical validation above them. The method is proven. The rural dataset is the work.

THE DIFFERENTIATOR

Dialect is a measurement problem. Vasl treats it as one.

A model that under-reads distress in African American Language — or in Appalachian English, or any other non-standard register — is not a slightly less accurate model. It fails hardest on the youth with the least access to care. Vasl's answer is a purpose-built corpus, a calibration method, and a parity constraint inside the estimator rather than in a report at the end.

CORPUS

VASL-AAVE-DISTRESS-001

The first clinically annotated corpus pairing dialect density with distress labels. No comparable dataset exists.

60,649 dialect-dense messages at a Blodgett model score of 0.80 or higher, drawn from a 1.1 million-message TwitterAAE frame.

Anchor-based seeding rather than instrument-based seeding, because instrument seeding reproduced the training prior instead of finding new language.

ANNOTATION

The community annotates

Primary raters bring both dialect competence and lived competence, with conflicts of interest disclosed on the record.

Independent community raters are paid and scored separately, so inter-rater agreement can be computed without the primary raters in the loop.

A clinical validation layer sits above community annotation, with agreement analysis by the data science team.

SDOH-aware calibration

Language variety is measured and carried as a covariate rather than ignored or corrected away, so social determinants do not silently absorb signal that belongs to distress. A provisional patent application covering the method was filed on September 3, 2026: *SDOH-Aware Behavioral Health Risk Scoring Using Measured Language-Variety Covariates*.

Parity inside the estimator

The subgroup parity constraint is part of estimation, not a fairness audit appended after training. Combined with slice-level release gating, that means a model cannot buy an aggregate accuracy gain by getting worse on the registers Vasl exists to read.

TERMINOLOGY, USED DELIBERATELY

Vasl says **Black** or **Black and African American** for people and communities, because the speaker population includes Black communities of Caribbean and other national origins. Vasl says **African American Language**, or AAVE, for the language variety, because that is the established linguistic term. Conflating the two is how research designs end up measuring the wrong thing.

Related work in progress: a pre-registered fairness audit of behavioral health language in ED clinical notes, using MIMIC under a credentialed PhysioNet data use agreement, targeted at *JAMIA*.

STATED PLAINLY

What is established, what is not.

Clinical buyers deserve a straight answer on maturity. Here is the current line between what Vasl has built and what Vasl has proven.

Established

- Five role-scoped portals and the VLAP service running in production on AWS
- Taxonomy frozen at VASL-VLAP-TAX-001 v1.0 — 47 signals, six categories
- VASL-AAVE-DISTRESS-001 corpus built and under annotation
- Provisional patent application filed September 3, 2026
- Community pilot outcomes: 79.5% 30-day retention, 42% PHQ-8 improvement at 90 days
- Internal evaluation of signal surfacing and escalation-rule handling on 106 synthetic cases

Not yet established

- Prospective clinical validation against C-SSRS is not complete
- No IRB-approved prospective study has been completed
- VLAP is not FDA-cleared and is not offered as a medical device
- Vasl holds no SOC 2, PCI DSS, or GDPR certification; SOC 2 Type II is a roadmap item, not a current attestation
- Performance figures to date come from internal evaluation and community program outcomes, not from a peer-reviewed prospective trial

The validation program under way

NIMH SBIR Phase I, submitted September 3, 2026. A six-month, \$581,149 study validating VLAP for suicide-risk signal detection in Black youth. Aim 1 is prospective validation in 180 youth aged 13 to 24, at least 90 of them Black, recruited across school-based health centers, FQHCs, and community behavioral health organizations, benchmarked against C-SSRS and against generic NLP, with pre-specified targets of 85% or better sensitivity and 80% or better specificity. Aim 2 integrates the engine into live workflow at school-based health centers.

Health-system data partnerships for model training and prospective validation are in negotiation, alongside separate validation of the suicide-risk use case against real clinical datasets already in hand.

START HERE

A retrospective read on your own notes

Phase 1 requires de-identified historical behavioral health notes and no change to anyone's workflow. It answers the only question that matters before a live deployment: how much distress language is sitting unread in your documentation, and how much of it is in registers your current tooling does not read.

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